Thread_ID

It is a unique session identified by a `thread_id`.
Every time, we invoke the LangGraph agent with the same `thread_id`, it is considered the same thread meaning that the agent remembers previous state via checkpoints.

So, to simply put:

- Same thread = same ongoing conversation (state is carried over)
- Different thread = fresh new start

Code Example:

```
from langgraph.checkpoint.memory import InMemorySaver
from langgraph.graph import StateGraph

checkpointer = InMemorySaver()
graph = compile(checkpointer=checkpointer)

# Thread A — first message
graph.invoke({"messages": ["Hello"]}, config={"configurable": {"thread_id": "thread-A"}})

# Thread A — second message (remembers the first)
graph.invoke({"messages": ["What did I just say?"]}, config={"configurable": {"thread_id": "thread-A"}})

# Thread B — completely fresh, knows nothing about Thread A
graph.invoke({"messages": ["Hello"]}, config={"configurable": {"thread_id": "thread-B"}})
```

The relationship between the `thread_id` and `StateBackend`

- The `StateBackend` stores files within a thread's state
- Cross-thread = a different `thread_id` -- the StateBackend files are not shared here
- Checkpointing = Save the thread state after each step so it can be resumed